Patient Education and Counseling 80 (2010) 212–218

Contents lists available at ScienceDirect

Patient Education and Counseling

journal homepage: www.elsevier.com/locate/pateducou

Intervention

Randomised controlled trial of a pedometer-based telephone intervention to increase physical activity among cardiac patients not attending cardiac rehabilitation

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ARTICLE INFO

Article history:
Received 28 May 2009
Received in revised form 8 October 2009
Accepted 6 November 2009

Keywords:
Physical activity
Pedometer
Cardiac rehabilitation
Randomised controlled trial

ABSTRACT

Objective: To determine the effectiveness of a pedometer-based telephone intervention on the physical activity levels of cardiac patients who did not attend a CRP.
Methods: A randomised controlled trial was conducted with 215 patients referred to a CRP but who could not or chose not to attend. The 6-week intervention included self-monitoring of physical activity using a pedometer and step calendar; and behavioural counselling and goal setting sessions. Data were collected at baseline, 6 weeks and 6 months.
Results: Study groups did not differ significantly at baseline. After 6 weeks, improvements in total physical activity time ($p = 0.027$), total physical activity sessions ($p = 0.003$), walking time ($p = 0.013$) and walking sessions ($p = 0.002$) in the intervention group were significantly greater than the control group after adjusting for baseline differences, and remained significant at 6 months.
Conclusion: The findings that the pedometer-based telephone intervention was successful in increasing physical activity levels in cardiac patients who did not attend a CRP could result in major health benefits for this group of people.
Practical implications: The pedometer-based telephone intervention could be offered as an effective and accessible option for patients not attending a CRP to increase and maintain their physical activity levels after hospitalisation.

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1. Introduction

Within Australia, approximately 70–80% of eligible patients do not attend a cardiac rehabilitation program (CRP) [1]. This low uptake rate of cardiac rehabilitation represents a public health problem as cardiac patients who do not attend a CRP are more likely to be physically inactive, have a higher risk-factor profile and poorer risk-factor knowledge [2], and are more socioeconomically disadvantaged [3] than those who do attend a CRP. There is clear evidence of the benefits of exercise-based cardiac rehabilitation in reducing cardiac mortality by 31% [4]. Physical activity interventions are a cost-effective approach for people with cardiovascular disease [5]. A randomised controlled trial of exercise training versus percutaneous coronary angioplasty in people with stable coronary artery disease found that the exercise training group had a higher event free survival rate and lower healthcare costs because of reduced hospitalizations and fewer repeat revascularizations [6].

There is evidence to suggest that community-based group programs are beneficial in reducing the effects of the chronic disease [7]. While community-based group programs may have a wider reach than hospital-based programs [8], the barriers to community-based group programs are likely to be similar to those reported by Scott et al. [1] that prevent cardiac patient's attendance at a CRP. Home-based interventions may reduce these barriers and be more applicable to a wider population of people with chronic disease. While home-based models with a physical activity component may be at least as beneficial as traditional hospital-based programs in improving physical activity levels [9], there is little research on physical activity interventions for cardiac patients who do not access a CRP. A recent systematic review reported positive outcomes for 69% of the telephone-based

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doi:10.1016/j.pec.2009.11.012

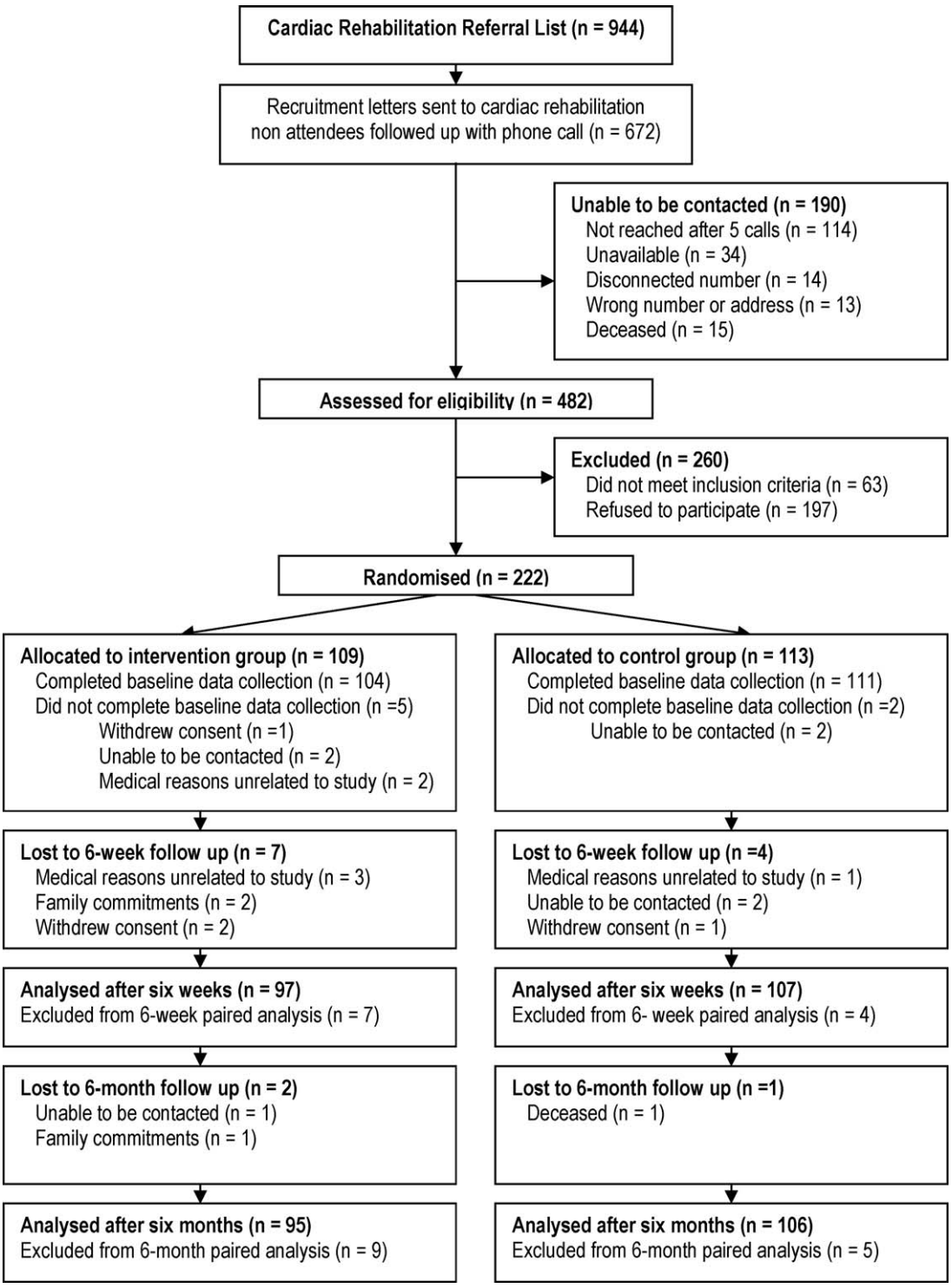


Fig. 1. Flow of participants through the trial.

physical activity intervention trials that had been conducted with healthy adults and those with chronic diseases; most interventions in these trials included multiple components, such as printed educational materials, pedometers, and telephone support [10]. Pedometers have gained acceptance as a motivational and self-monitoring tool for increasing physical activity levels in healthy individuals and in those with chronic illnesses [11] and have been used with behavioural interventions based on Bandura's [12] social cognitive theory. The aim of this study was to determine the effectiveness of an individualised pedometer-based telephone intervention on physical activity levels of cardiac patients who were referred to, but could not, or chose not to attend, a CRP.

2. Methods

2.1. Study design and study population

A randomised controlled trial was used (Fig. 1). Cardiac patients who were referred to an outpatient CRP in the Illawarra or Shoalhaven area of New South Wales, Australia, over a 10-month

period during 2005–2006, were identified from a centralised referral database maintained by the area health service. Patients were eligible for this study if they had not attended an outpatient CRP, spoke English and had no co-morbidities affecting physical activity participation; they were invited to participate 2 months after discharge from hospital.

2.2. Randomisation and blinding

A set of 250 random numbers was generated using Microsoft Excel and the Statistical Package for Interactive Data Analysis (SPIDA) was used to group these numbers by blocks of four to ensure a balanced sample size across both study groups. The random numbers were generated and packaged into sealed envelopes by a researcher external to the study. Group allocation was concealed from the researcher until the participant had agreed to be in the study but was not concealed from the researcher when delivering the intervention or assessing outcomes. Potential participants were invited to take part in The Illawarra and Shoalhaven Healthy Heart Project. Participants were not told whether they had been allocated to the intervention or control group and remained blind to group allocation. When a patient had verbally agreed to enrol into the study they were sent additional information and an informed consent form to sign and return to the researcher.

2.3. Data collection and measurement instruments

Data on physical activity, psychosocial factors affecting physical activity, and demographic information were collected at baseline, 6 weeks and 6 months via a telephone questionnaire. The main outcome measure was total physical activity per week. The Active Australia Questionnaire [13] which has demonstrated reliability in Australian populations [14] was used to assess self-reported physical activity.

Psychosocial status was a secondary outcome. Data on self-efficacy for exercise, was collected using the self-efficacy for exercise scale (adapted from Resnick and Jenkins [15]). Outcome expectancies for participation in physical activity were assessed using a scale adapted from Kobau and DiIorio's [16] research. Self-management strategy use was assessed using behavioural and cognitive strategy scales adapted from Saelens et al. [17]. Psychological distress was assessed using a Kessler 6 (K-6) scale adapted from the NSW Health Older Peoples Survey [18]. The questionnaire was pilot tested with a sample of cardiac patients and psychosocial measures demonstrated acceptable internal consistency values (self-efficacy Cronbach's alpha (α) = 0.92; outcome expectancies α = 0.80; self-management strategies α = 0.80; K-6 α = 0.85).

2.4. Intervention

The intervention was based on social cognitive theory [12]. The intervention strategies included a focus on increasing self-efficacy, increasing beliefs about the positive health consequences of taking action (outcome expectancies) and establishing physical activity goals [19]. The intervention comprised a pedometer, a step calendar for self-monitoring, and telephone support which included goal setting and behavioural reinforcement. The pedometer was a Yamax Digiwalker 700B, a model that has been found to be valid and reliable [20]. Goal setting varied from people nominating specific step count targets or a time-based target. The behavioural counselling advice focused on increasing confidence, supplemented by feedback from the pedometer. Participants were asked to develop strategies to overcome barriers to increasing their physical activity.

Interactions with participants were carried out by mail and telephone. Intervention participants were instructed not to change their physical activity patterns to allow a baseline step count to be obtained. Following the baseline data collection period, participants were instructed to record pedometer steps per day for 6 weeks. Behavioural counselling and goal setting sessions were provided via telephone (about 15 min) to intervention participants 1 week after they had received the pedometer and step calendar, and at the third week. After a 6 six-week intervention period, participants received "booster" telephone calls at weeks 12 and 18; these calls were designed to offer feedback on goal achievement and support.

The control group was sent two physical activity information brochures by mail after they had completed the baseline questionnaire. The same brochures were mailed to the intervention group participants. The control group participants did not receive any reinforcement telephone calls between baseline and the 6-week and 6-month follow-up questionnaires.

2.5. Statistical analyses

Sample size calculation was based on the number of sessions of total physical activity per week. To detect a difference of 2.5 sessions per week between the intervention and control group, a sample size of 91 in each group was required. To account for up to 20% loss to follow-up, the sample was increased to 108 per group. Paired analyses with complete cases were performed using SPSS 15.0 (SPSS Inc., Chicago, IL) and two tailed p values <0.05 were considered significant. Independent t-tests and Mann–Whitney U tests were used to compare continuous variables between groups. Paired t-tests and Wilcoxon rank tests were used to detect changes in continuous variables within groups. For categorical variables, Pearson's Chi-square test was used to detect changes between groups and McNemar's Chi-squared test was used to detect changes within groups. An analysis of variance (ANOVA) was conducted to compare the change variables within groups and determine any differences between the intervention and the control group, adjusting for baseline differences. Cohen's effect sizes for between group differences in physical activity variables were calculated by dividing the change between the intervention and control groups by the pooled standard deviation [21].

The study was approved by the Human Research Ethics Committees of the University of Wollongong/Illawarra Area Health Service and the University of New South Wales.

3. Results

3.1. Response rate and baseline characteristics

There were 672 cardiac patients who were referred to, but did not attend a CR program during the 10-month recruitment period. Of these, 482 (71.7%) were contacted and assessed for eligibility into the trial. Sixty-three did not meet the inclusion criteria and 197 refused to participate in the trial. The demographic characteristics of participants were compared with those who were excluded (i.e., did not consent to participate or were not eligible to participate) and with those who could not be reached to identify any differences between these groups (data not shown). Participants were 3.4 year younger than those who were excluded (95% CI: −5.4 to −1.4) and 2.3 years older than those who were not reached (95% CI: 0.1–4.6). The proportion of male participants relative to female participants was significantly greater than the proportion that were excluded (Δ = 8.9%; 95% CI: 0.5–17.2), however, there was no difference between the proportion of male participants

relative to female participants and those not reached. Participants were significantly less disadvantaged than those who were excluded ($\Delta = -9.6\%$; 95% CI: −16.7 to −2.4) and those who were not able to be reached ($\Delta = -8.5\%$; 95% CI: −16.3 to −0.7). There were no significant differences between participants and those excluded or not reached for cardiac disease diagnosis.

A total of 222 (46.1% of those contacted) consented to participate in the trial and were randomly allocated into the intervention or control group. Two hundred and fifteen participants completed the baseline questionnaire. Table 1 shows no significant differences in baseline characteristics between the study groups.

The response rates for the 6-week and 6-month questionnaires were 94.9 and 93.5%, respectively. Fig. 1 details the flow of participants through the trial.

Table 1
Baseline sociodemographic and clinical characteristics of participants.

Characteristic	Intervention Group *N* (%)	Control Group *N* (%)	*p* value (x^2)
Number in study group	104 (48.4)	111 (51.6)	–
Age, mean years (SD)	66.7 (10.6)	65.4 (11.5)	0.394[a]
Gender			
Male	75 (71.2)	76 (68.5)	0.559
Female	29 (27.9)	35 (31.5)	
Body mass index[b] (SD)	27.3 (5.2)	27.8 (4.8)	0.555[a]
Born in Australia	83 (79.8)	76 (68.5)	0.058
Marital status			
Married/de facto	77 (74.0)	82 (73.9)	
Widowed	9 (8.7)	17 (15.3)	0.308
Separated/divorced	12 (11.5)	8 (7.2)	
Never married	6 (5.8)	4 (3.6)	
Education			
Secondary level or less	70 (67.3)	70 (63.1)	0.513
Tertiary level	34 (32.7)	41 (36.9)	
Occupation			
Retired	70 (67.3)	74 (66.7)	
Currently employed	22 (21.2)	24 (21.6)	0.995
Unemployed/disability pension/volunteer/ sick leave	12 (11.5)	13 (11.7)	
Diagnosis[c]			
CABGS	15 (14.4)	21 (18.9)	
PCI	23 (22.1)	28 (25.2)	
MI	30 (28.9)	21 (18.9)	0.523
ACS	20 (19.2)	22 (19.8)	
Other	16 (15.4)	19 (17.2)	
Self-reported health status			
Excellent/very good	40 (38.5)	44 (39.6)	
Good	41 (39.4)	31 (27.9)	0.121
Fair/poor	23 (22.1)	36 (32.5)	
Smoking status			
Yes	12 (11.5)	9 (8.1)	
Not now but previously	54 (51.9)	63 (56.8)	0.876
Never	37 (35.6)	39 (35.1)	
Following dietary guidelines			
Yes	78 (75.0)	95 (85.6)	0.050
Sufficiently active			
≥150 min[d]	60 (57.7)	61 (55.0)	0.686
≥150 min and ≥5 sessions[e]	55 (52.9)	53 (47.7)	0.452

[a] Independent *t*-test.
[b] Body mass index = kg/m^2.
[c] CABGS, coronary artery bypass graft surgery; PCI, percutaneous coronary intervention; MI, myocardial infarction; ACS, acute coronary syndrome.
[d] Sufficiently active measured by achievement of ≥150 min total activity per week.
[e] Sufficiently active measured by achievement of ≥150 min total activity and ≥5 sessions of activity per week.

3.2. Physical activity

Within the intervention group, improvements in self-reported physical activity were highly significant from baseline to 6 weeks for total physical activity and walking variables (Table 2). Improvements were significantly greater in the intervention group than the control group for total physical activity minutes ($p = 0.027$, effect size = 0.31), total physical activity sessions ($p = 0.003$, effect size = 0.41), walking minutes ($p = 0.013$, effect size = 0.35) and walking sessions ($p = 0.002$, effect size = 0.43) after adjusting for baseline differences (data not shown in table). At 6 months, physical activity improvements from baseline remained significant within the intervention group with no change in controls (Table 2). Improvements were significantly greater in the intervention group than in the control group for total physical activity minutes ($p = 0.015$, effect size = 0.35), total physical activity sessions ($p = 0.019$, effect size = 0.32), walking minutes ($p = 0.002$, effect size = 0.45) and walking sessions ($p = 0.026$, effect size = 0.33) after adjusting for baseline differences (data not shown in table).

3.3. Psychosocial status

There were significant improvements in self-efficacy, outcome expectancies, and cognitive and behavioural self-management strategy use within the intervention group at 6 weeks, with the increase in self-efficacy and outcome expectancies remaining significant at 6 months. There was a significant decrease in psychological distress within the control group at 6 weeks and 6 months, however, there was no change in psychological distress in the intervention group (Table 3). At 6 weeks, the change in the intervention group was significantly greater than the change in the control group in outcome expectancies ($p = 0.038$; effect size = 0.30) and cognitive self-management strategy use ($p = 0.028$; effect size = 0.30) after adjusting for baseline differences (data not shown in table). At 6 months, there were no significant differences between groups after adjusting for baseline differences (data not shown in table).

4. Discussion and conclusion

4.1. Discussion

There are scant studies on physical activity interventions for cardiac patients who do not attend a CRP. The present study's findings that a pedometer-based telephone intervention was successful in increasing physical activity levels in cardiac patients who did not attend a CRP could result in a reduced risk of subsequent cardiovascular disease [22]. Redfern et al. [23] evaluated an intervention aimed at improving cardiovascular risk factors of acute coronary syndrome survivors who did not attend a CRP; their study reported that an intervention comprising a 1 h face-to-face consultation and four 10-min follow-up telephone calls over 3 months resulted in an improvement in physical activity levels over 1 year. The positive outcomes of our study and the one by Redfern et al. [23] show the potential for low-intensity, low-cost and community-based interventions for increasing physical activity among people who did not attend a CRP.

The intervention in the current study could be classified as "home based", as participants were able to engage in physical activity in their own time and local environments. The fact that almost half of cardiac rehabilitation non-attendees contacted for recruitment into our study agreed to participate in this intervention indicates that this population was willing to try alternatives to formal cardiac rehabilitation, and that this type of intervention was appealing. Other studies have demonstrated

Table 2
Self-reported physical activity changes at 6 weeks and 6 months.

	Intervention $n=97$				Control $n=107$			
	Baseline[a]	Six weeks	Δ	p value	Baseline	Six weeks	Δ	p value
6-Week changes								
Total physical activity								
Mean minutes (SD)	271.8 (238.4)	366.5 (270.8)	94.7 (213.9)	<0.001**,c	239.7 (229.6)	270.9 (244.4)	31.2 (193.5)	0.099
Median minutes[b]	210.0	300.0	90.0	<0.001**	180.0	190.0	10.0	0.072
Mean sessions (SD)	6.6 (4.5)	9.0 (5.7)	2.4 (6.2)	<0.001**,c	6.9 (5.6)	7.1 (5.6)	0.2 (4.4)	0.696
Median sessions	6.0	8.0	2.0	<0.001**	6.0	6.0	0.0	0.696
Walking								
Mean minutes (SD)	169.8 (165.3)	249.9 (196.0)	80.1 (176.1)	<0.001**,c	180.5 (164.7)	202.6 (189.5)	22.0 (154.0)	0.142
Median minutes	120.0	210.0	80.0	<0.001**	147.0	150.0	0.0	0.233
Mean sessions (SD)	5.3 (4.1)	7.2 (5.0)	1.9 (5.5)	0.001**,c	5.6 (3.9)	5.5 (4.0)	−0.1 (3.5)	0.824
Median sessions	5.0	7.0	2.0	0.001**	5.0	5.0	0.0	0.774
	Intervention $n=95$				Control $n=106$			
	Baseline[a]	Six months	Δ	p value	Baseline	Six months	Δ	p value
6-Month changes								
Total physical activity								
Mean minutes (SD)	272.3 (240.2)	358.7 (260.0)	86.4 (225.3)	<0.001**,c	241.4 (230.0)	250.9 (250.6)	9.4 (218.5)	0.658
Median minutes	210.0	315.0	91.0	<0.001**	180.0	150.0	0.0	0.965
Mean sessions (SD)	6.6 (4.5)	8.6 (5.7)	1.9 (5.4)	0.001**,c	6.9 (5.6)	7.0 (6.4)	0.1 (5.8)	0.920
Median sessions	6.0	7.0	1.0	0.001**	6.5	5.0	0.0	0.667
Walking								
Mean minutes (SD)	168.9 (166.4)	231.5 (189.1)	62.5 (172.3)	0.001**,c	181.7 (165.1)	168.3 (171.8)	−13.7 (167.3)	0.397
Median minutes	120.0	200.0	60.0	<0.001**	148.5	120.0	0.0	0.325
Mean sessions (SD)	5.4 (4.1)	6.8 (5.0)	1.4 (4.8)	0.006**,c	5.6 (3.9)	5.5 (4.9)	−0.2 (5.0)	0.741
Median sessions	5.0	6.0	1.0	0.015*	5.0	4.0	0.0	0.197

[a] Baseline data compared to 6 weeks is different to baseline data compared to 6 months as values are paired for analysis.
[b] The median change is not necessarily equal to the mathematical difference between the median baseline value and the median 6-week value because the median baseline, 6 weeks, and change values represent the midpoints of three distinct distributions.
[c] Change in the intervention group was significantly different to change in control group at $p < 0.05$ after adjusting for baseline differences (only applicable to mean values, not median).
* $p < 0.05$.
** $p < 0.01$.

the success of home-based programs in increasing physical activity levels of cardiac patients [23,24].

The present study's findings that participants within the intervention group had enhanced self-efficacy for exercise at 6 weeks and 6 months differs from the findings of the study using a similar pedometer-based intervention conducted with people who had attended a CRP which reported no change in self-efficacy for exercise; the authors speculated that participant's recent attendance at a CRP may have increased their confidence to undertake physical activity before they took part in the study [24]. Outcome expectancies related to physical activity participation increased significantly in the intervention group compared with the control group at 6 weeks in our study, however, no significant differences between groups were observed at 6 months, indicating that the effect of the intervention on outcome expectancies was not as strong at 6 months as at 6 weeks. Associations between positive outcome expectations and increased participation in physical activities have been described [25] and are supported by the present study's findings as the improvement in outcome expectancies in the intervention group at 6 weeks occurred concurrently with an increase in self-reported physical activity. Similar findings were reported by Butler et al. [24].

There was a significant increase in the use of behavioural and cognitive self-management strategies in the intervention group at 6 weeks, but not at 6 months. The lack of self-management strategy use at 6 months may have been because changes in physical activity had already been made at 6 weeks and participants may not have been consciously aware of these strategies at 6 months. In populations living with a chronic illness, self-management interventions have been developed [26], however, to our knowledge the study by Butler et al. [24] is the only other study to have investigated the efficacy of teaching self-management strategies to increase physical activity among those in the community living with cardiovascular disease; that study also reported positive findings at 6 weeks, however, there was also a significant improvement in cognitive self-management strategy use at 6 months.

There was a significant decrease in psychological distress in the control group, but not in the intervention group; however, there were no significant differences between the two study groups at any time point. While a decline in psychological distress due to the recovery process could be expected in all participants regardless of group allocation, we did not expect to see a decline in the control group in isolation. We expected that the intervention group would have a lower level of psychological distress due to their increased participation in physical activity [27]. While we have no plausible explanation for this finding, further analysis of this result is warranted.

A limitation of this study includes a possible self-selection bias due to volunteer enrolment as participants may have been a more motivated group than those who did not participate. However, this would have applied equally to controls, and still we observed significant intervention effects. Generalisability of the findings may be affected due to some of the differences in age, gender and socioeconomic status of participants compared with those who were excluded and those who could not be reached. While the possible effect of bias using a self-reported physical activity assessment tool could not be dismissed, a randomised control trial conducted by the investigators during the same period using objective measures of cardiorespiratory fitness with patients attending a CRP found that self-reported physical activity changes using the Active Australia Questionnaire were corroborated by improvements in cardiorespiratory fitness in the intervention group [24].

Table 3
Self-reported changes in psychosocial status at 6 weeks and 6 months.

	Intervention $n=97$				Control $n=107$			
	Baseline	Six weeks	Δ	p value	Baseline	Six weeks	Δ	p value
6-Week changes								
SEE[a]								
Mean (SD)	4.53 (1.34)	4.95 (1.2)	0.38 (1.2)	<0.001**	4.50 (1.49)	4.65 (1.4)	0.15 (1.1)	0.178
Median	4.88	5.13	0.25	0.001**	4.75	4.75	0.0	0.330
Outcome expectancies[b]								
Mean (SD)	3.72 (0.61)	3.91 (0.5)	0.19 (0.5)	0.001**,e	3.75 (0.64)	3.78 (0.6)	0.04 (0.5)	0.458
Median	3.75	4.00	0.13	0.001**	3.75	3.88	0.13	0.249
Self-management[c]								
Behavioural								
Mean (SD)	2.58 (0.87)	2.76 (0.9)	0.17 (0.7)	0.022*	2.61 (0.92)	2.74 (0.9)	0.13 (0.7)	0.056
Median	2.75	2.75	0.25	0.030*	2.50	2.75	0.0	0.060
Cognitive								
Mean (SD)	3.58 (1.03)	3.85 (1.0)	0.27 (0.9)	0.003**,e	3.70 (1.00)	3.73 (1.0)	0.03 (0.7)	0.714
Median	3.75	4.00	0.25	<0.001**	4.00	4.00	0.0	0.478
Psychological distress (K6)[d]								
Mean (SD)	10.11 (4.59)	9.52 (4.2)	−0.60 (3.1)	0.064	10.67 (4.70)	9.78 (4.5)	−0.91 (4.0)	0.021*
Median	8.00	8.00	0.0	0.050	9.00	8.00	0.0	0.008*
	Intervention $n=95$				Control $n=106$			
	Baseline	Six months	Δ	p value	Baseline	Six months	Δ	p value
6-Month changes								
SEE[a]								
Mean (SD)	4.54 (1.34)	5.00 (1.2)	0.45 (1.3)	0.001**	4.52 (1.34)	4.62 (1.5)	0.10 (1.3)	0.422
Median	4.88	5.25	0.38	0.001**	4.88	4.69	0.0	0.536
Outcome expectancies[b]								
Mean (SD)	3.73 (0.61)	3.84 (0.5)	0.11 (0.5)	0.032*	3.73 (0.61)	3.75 (0.7)	0.00 (0.5)	0.962
Median	3.75	4.00	0.13	0.036*	3.75	3.88	0.0	0.682
Self-management[c]								
Behavioural								
Mean (SD)	2.58 (0.88)	2.61 (0.9)	0.03 (0.9)	0.764	2.58 (0.89)	2.50 (0.9)	−0.11 (0.9)	0.185
Median	2.75	2.50	0.00	0.787	2.75	2.50	0.0	0.313
Cognitive								
Mean (SD)	3.57 (1.04)	3.76 (1.0)	0.20 (1.1)	0.075	3.57 (1.04)	3.74 (1.1)	0.04 (1.0)	0.686
Median	3.75	4.00	0.25	0.076	3.75	4.00	0.25	0.475
Kessler 6 psychological distress[d]								
Mean (SD)	10.13 (4.62)	9.58 (4.4)	−0.55 (3.7)	0.156	10.13 (4.62)	9.67 (4.5)	−0.98 (3.8)	0.009**
Median	8.00	8.00	0.0	0.101	8.00	8.00	0.0	0.005**

[a] SEE, self-efficacy for exercise scale (adapted from Resnick and Jenkins [15]), higher score = greater self-efficacy.
[b] Outcome expectancies scale (adapted from Kobau and Dilorio [16]), higher score = more positive outcomes.
[c] Self-management scale (adapted from Saelens et al. [17]), higher score = greater self-management strategy use.
[d] Kessler 6 Psychological distress scale (NSW Health Department [18]), higher score = greater psychological distress.
[e] Change in the intervention group was significantly different to change in control group at $p < 0.05$ after adjusting for baseline differences.
* $p < 0.05$.
** $p < 0.01$.

4.2. Conclusion

The findings that the pedometer-based telephone intervention was successful in increasing physical activity levels in cardiac patients who chose not to attend a CRP could result in major health benefits for this group of people. This study contributes significantly to the evidence base for 'best practice' interventions to support people with cardiovascular disease to be more physically active, thus having the potential to reduce the cardiovascular disease burden within the community.

4.3. Practical implications

The pedometer-based intervention could be offered as an effective and accessible option for people not attending a cardiac rehabilitation program to increase and maintain their physical activity levels after hospitalisation.

Conflict of interest

None.

Acknowledgements

This project was funded through the NSW Health Promotion Research Demonstration Grant Scheme, 2004/2005. The authors wish to thank the study participants; the contributing cardiologists, exercise physiologists, and cardiac rehabilitation staff in the Illawarra and Shoalhaven regions of South Eastern Sydney & Illawarra Area Health Service; the Steering Committee; and Tania Starr and Darren Mayne.

Funding: NSW Health Promotion Research Demonstration Grant Scheme.

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